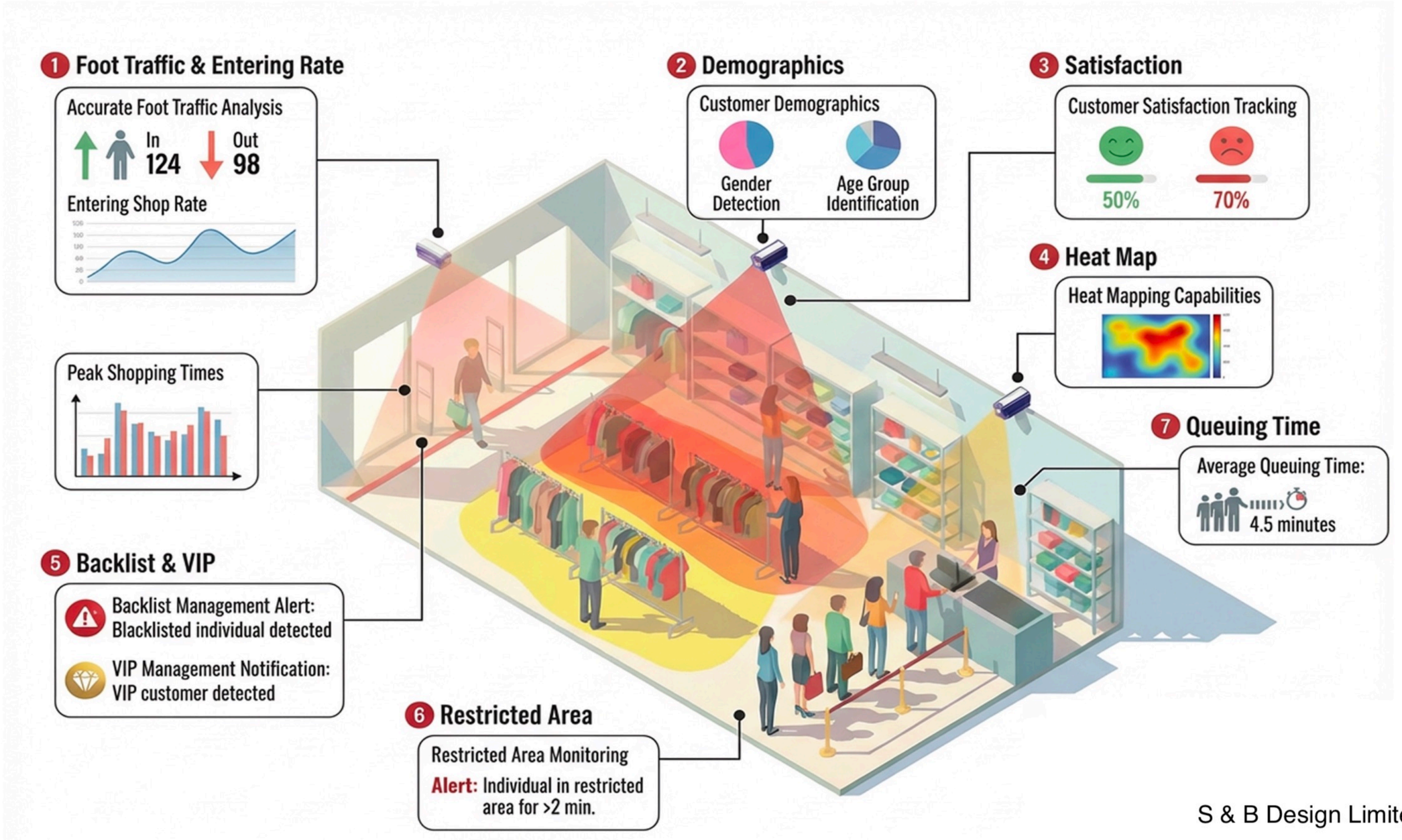
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People counting
In Retail
Data & Analytics



People counting by AI Camera

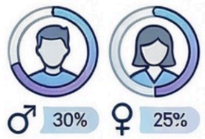


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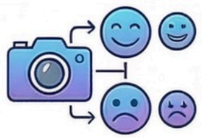
People counting by AI Camera



Accurate Foot Traffic Analysis
- Real-time data on the number of customers entering and exiting the store.
- Peak shopping times visualized graphically.



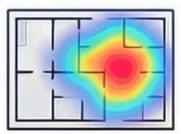
Customer Demographics
- **Gender Detection**: Gender distribution statistics presented on the dashboard.
- **Age Group Identification**: Age group segmentation displayed for quick insights.



Customer Satisfaction Tracking
- Analyzes customer facial expressions to gauge satisfaction (e.g., happy, unhappy).



Entering Shop Rate
- Measures the rate at which customers enter the store.



Heat Mapping Capabilities
- Visualization of customer traffic patterns in popular zones.



Backlist Management
- Alerts displayed when backlisted individuals enter the premises.



VIP Management
- Notifications when VIP customers are recognized.



Restricted Area Monitoring
- Immediate alerts if individuals stay in restricted areas for specified durations.

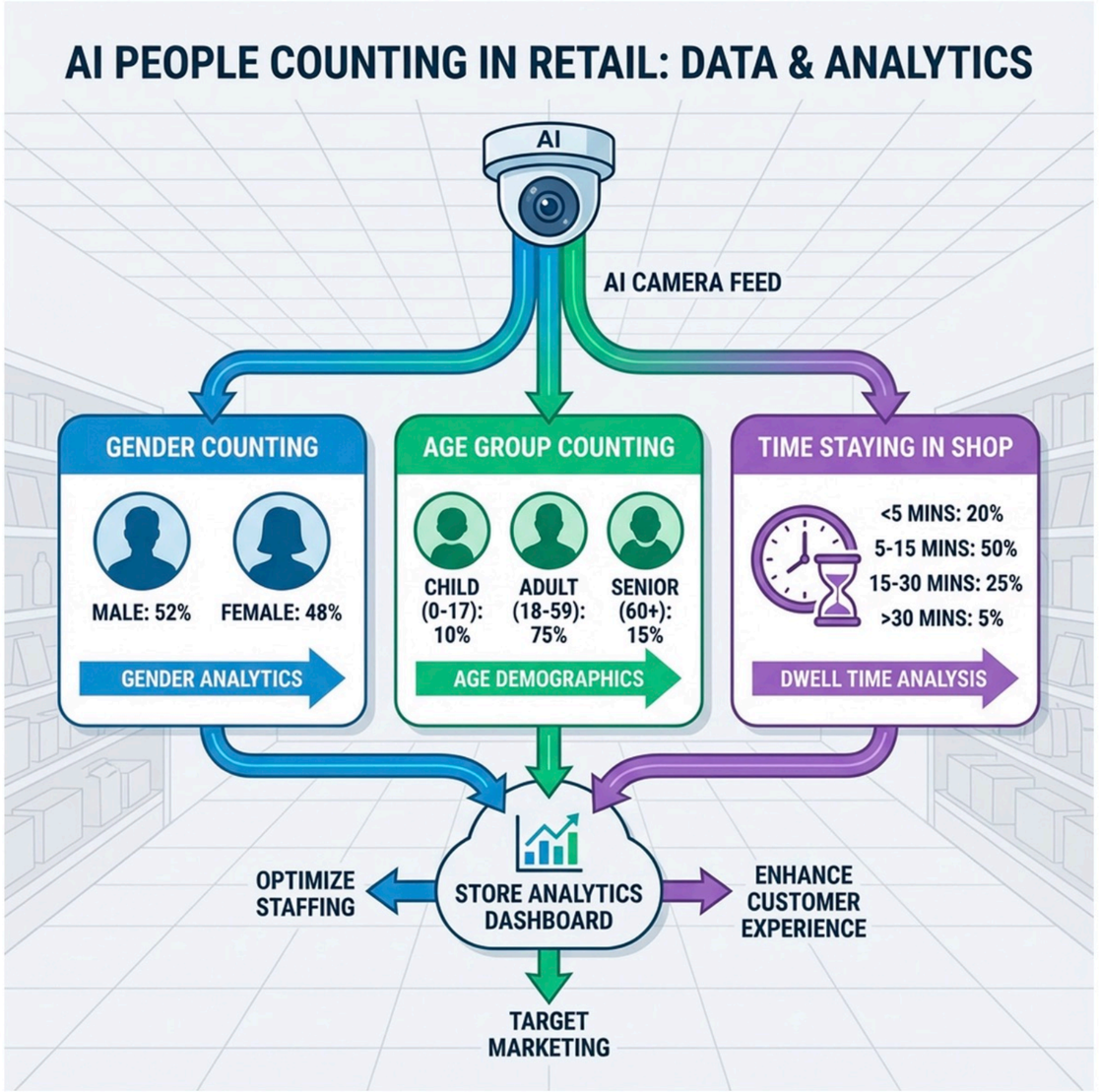


Average Queuing Time
- Displays the average wait time in the cashier line for customer awareness.



Language Support
- Dashboard available in multiple languages (Chinese and English).

People counting - Data Analysis





Customer Behavior Insights
Requires analysis of movement patterns for insights into store layout effectiveness.



Operational Efficiency
Analysis based on foot traffic data to optimize staffing levels.



Sales Conversion Rates
Correlates foot traffic data with sales performance for analysis.



Real-Time Data Analysis
Comparative data over time (e.g., same period last week) for trend identification.

People counting - Data Analysis



Integration with Other IoT Devices
Custom data experiences based on combined sensor input.



Data Visualization Tools
Customizable reports based on comprehensive data analysis.



Actionable Insights
Derived from aggregated data for marketing strategies and sales enhancements.

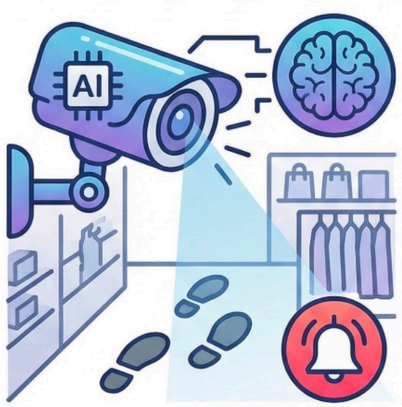


Scalability
Assess proportional data as it scales with the retail environment.

Dashboard Screenshot



People counting - Security Features



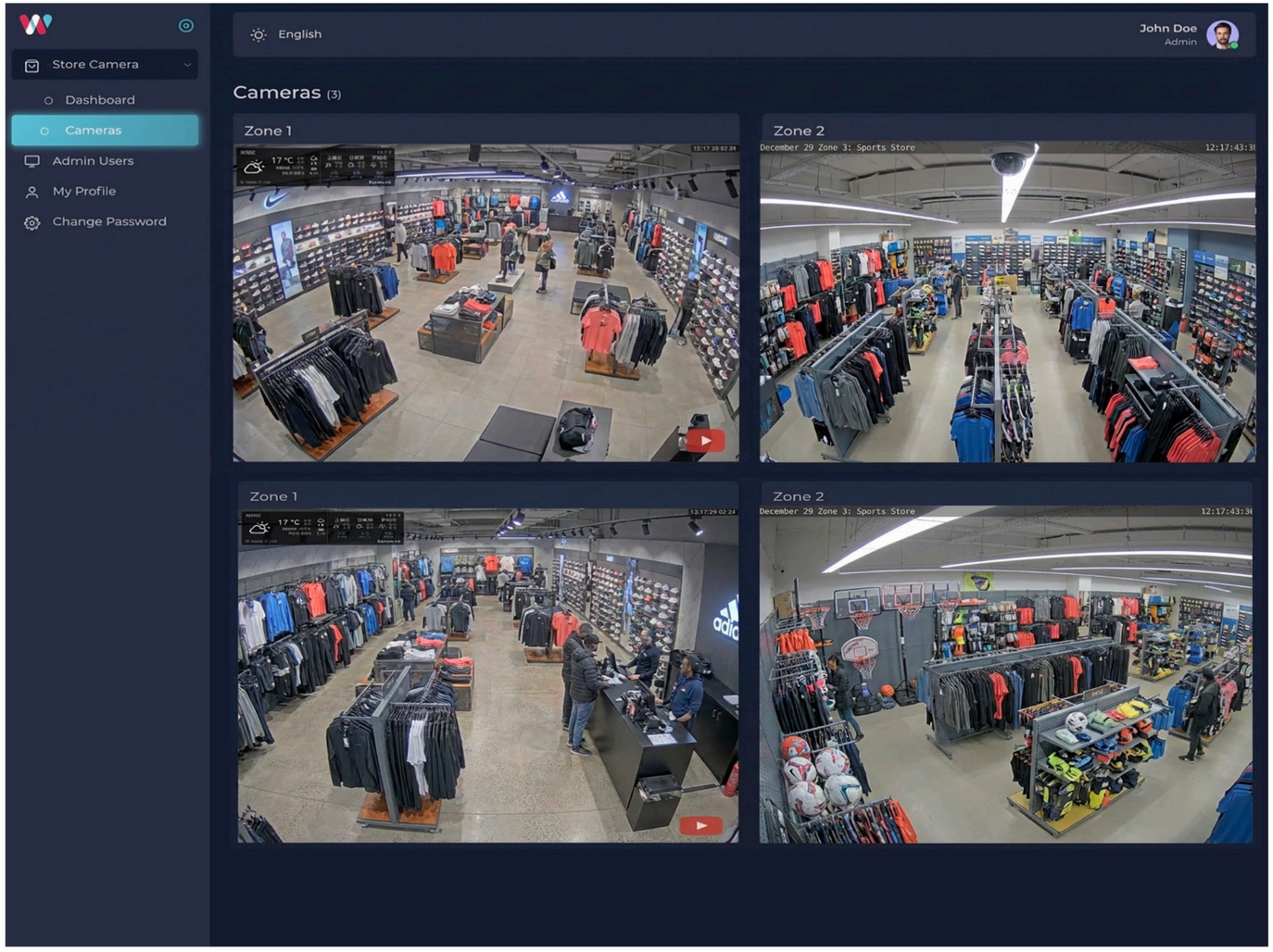
Security Monitoring

- Utilizes AI camera technology to monitor and detect suspicious activities in real time.
- Captures activity in high-risk areas, providing alerts for theft, vandalism, or unauthorized access.



After-Hours Monitoring

- Cameras continue to operate when the shop is closed, ensuring constant surveillance.
- Alerts for any unauthorized access or suspicious activities during non-business hours, enhancing the security of the premises.



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Asset Management (Indoor positioning) / AI Camera Solution

Use case for Museum



Asset Control and Security System



1. Too Close Reminder
The AI camera monitors visitor proximity to exhibits. If visitors stand too close, the system triggers the LoRaWAN buzzer to broadcast a reminder for them to step back.

2. Visitor Counting
Accurately tracks the number of visitors entering and exiting the museum, providing valuable attendance statistics.

3. Demographic Analysis
Analyzes visitor demographics, including age and gender, to gain insights into the audience profile.

4. Suspicious Object Detection
Monitors designated areas for suspicious objects, automatically triggering alerts for staff intervention when detected.

Asset Control and Security System



5. Artwork and Object Removal Monitoring
Detects any unauthorized removal of artwork or objects, sending immediate alerts to security personnel for prompt action.



6. Exhibit Interaction Analysis
Tracks the duration visitors spend at individual exhibits, helping to evaluate engagement levels and optimize displays.



7. Crowd Management
Provides real-time data on crowd density, enabling effective management of visitor flow and enhancing overall safety.



8. Facial Expression Analysis
Gauges visitor satisfaction and engagement by analyzing facial expressions while they interact with exhibits.



9. Instant Removal Alert
A Time-of-Flight (ToF) distance sensor, placed behind exhibits, triggers an alarm if any exhibit is removed.

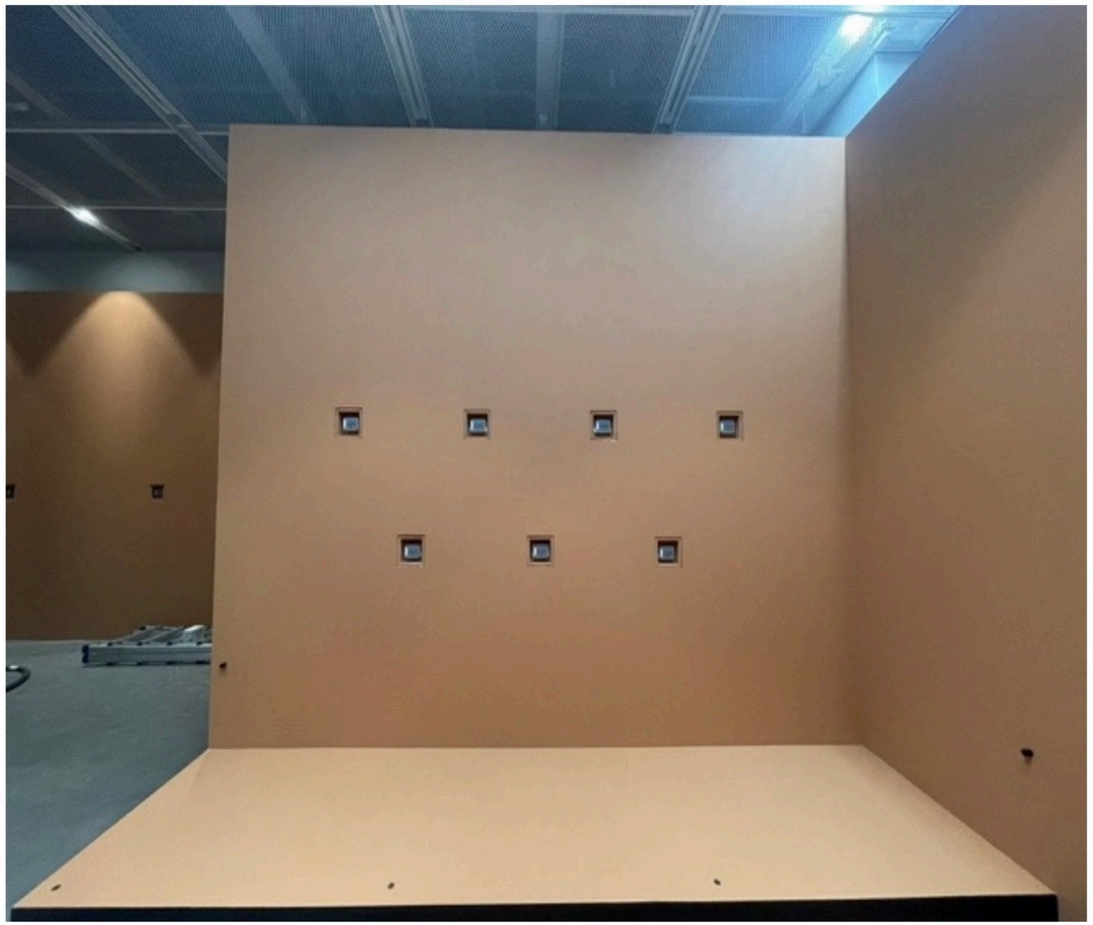


10. Asset Management
An Angle of Arrival (AoA)-supported asset beacon is attached to each artwork, facilitating easy location management for efficient asset tracking.

TOF Sensor Beacon

The TOF sensor detects the presence of artworks in real-time, ensuring that any removal triggers an immediate alarm. This advanced distance measurement sensor enhances security by monitoring the status of the artworks, allowing for prompt response to any unauthorized actions.



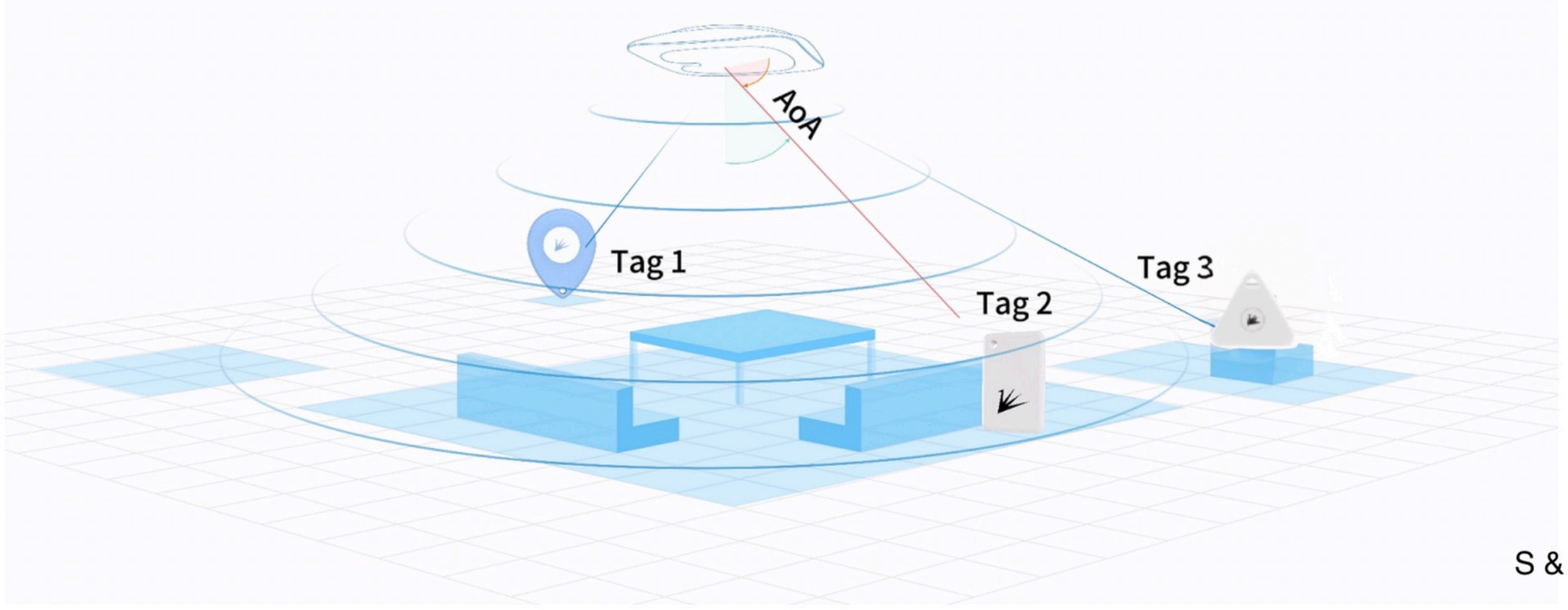


Asset Control with indoor positioning

Location Artworks real-time location:

- The AoA locator collects data from BLE AoA beacons attached to artworks, which transmit Bluetooth signals containing specific information. It utilizes an integrated receiving antenna array to determine the direction of these signals. Additionally, the AoA locator employs triangulation technology to pinpoint its own location and uploads the data to the Position Engine, facilitating efficient asset management for artworks in the museum.

- By utilizing the same AoA technology, we will install AoA gateways at each entrance and exit of the museum. This setup will trigger an alarm if any artwork is detected in those areas, thereby preventing potential theft.



BLE Sensors and Gateway



ToF Distance Sensor

- Impressive range of up to 3 meters
- High precision millimeter-level ranging accuracy
- Exceptional 3-year battery life for long-lasting usage
- IP67 waterproof rating for protection against water and dust
- Measures distance using time-of-flight (ToF) technology



Asset Tag

- Battery can be replaced when it is exhausted.
- Pre-programmed with standard beacon firmware.
- Up to 10 slots of advertising frames.
- Supports iBeacon, Eddystone (e.g., URL, UID, TLM), and AltBeacon.
- Advertising range up to 400m in open areas.



Indoor AOA Locator

- Accuracy : 1meter
- Dual System
- PoE/DC Power Supply
- Ethernet
- IP66
- Bluetooth Low Energy BLE AoA Anchor GA25

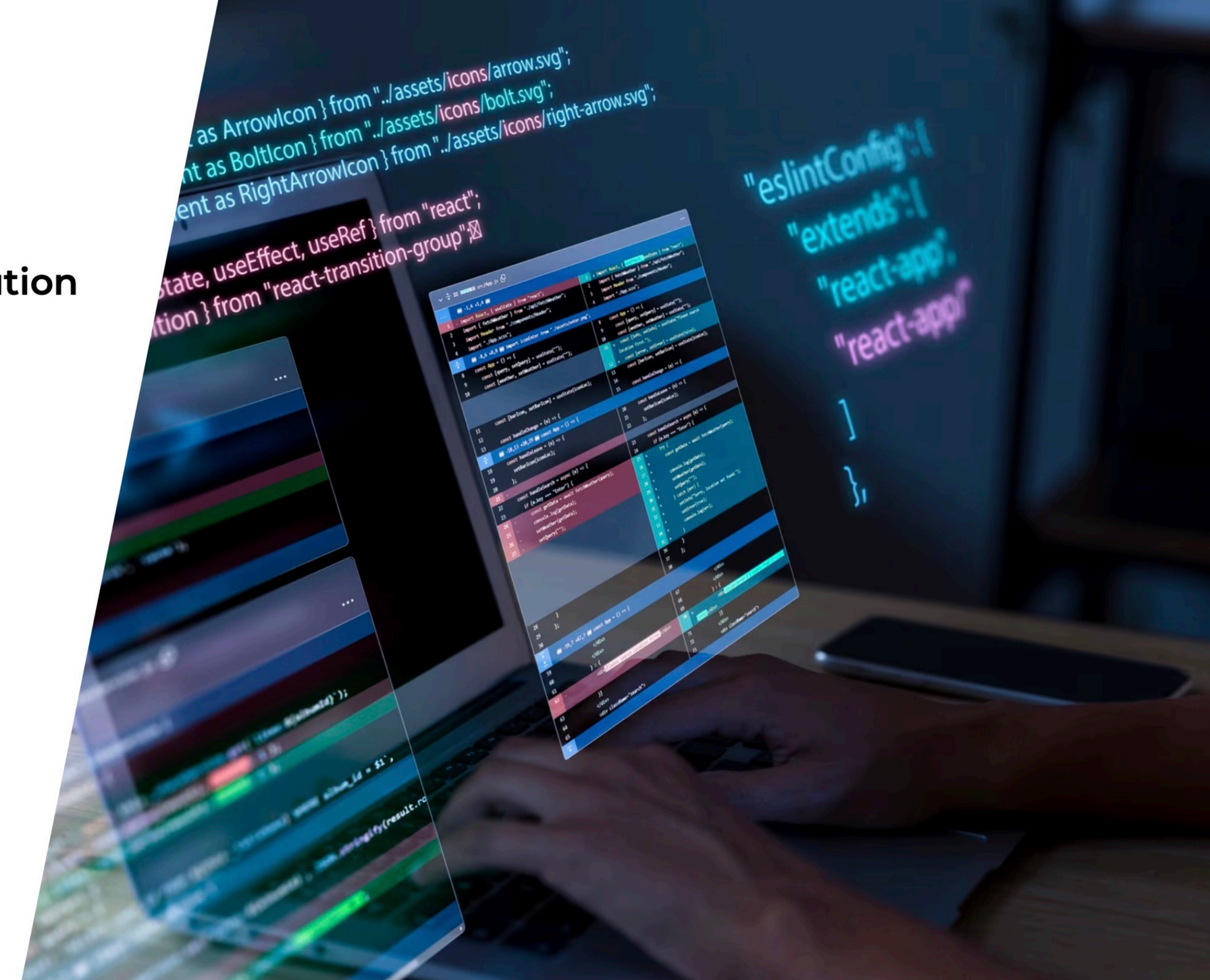


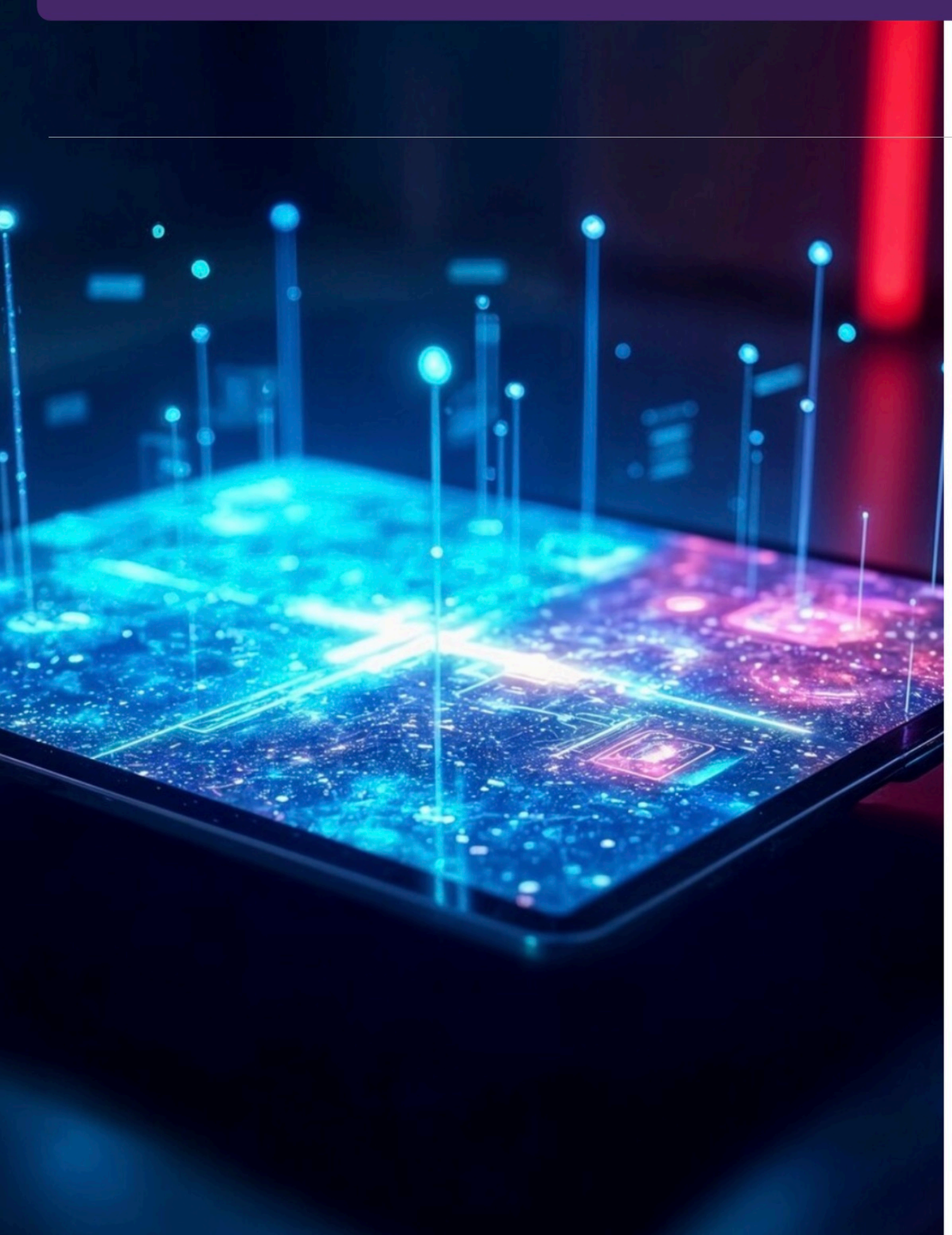
4MP IR Bullet WizMind Network Camera

Need a customized solution for your industry?

As experienced software developers, we tailor our platform to meet your specific requirements.

Contact us to discuss your needs.





Contact Us

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